Raghavendra Kumar Sakali

Assistant Professor
Department of Computer Science and Engineering
Shiv Nadar University Chennai

Chennai – 603110 | 8074193581, 9490064573
rksakali.it29@gmail.com

Professional Profile

Assistant Professor in Computer Science and Engineering with experience in teaching, research, curriculum delivery, student mentoring, and academic service. Holds a Ph.D. in Computer Science and Engineering, specializing in fault-tolerant adder and multiplier architectures for mission-critical systems. Research interests include reliable and approximate arithmetic circuits, FPGA-based computing, and hardware acceleration for image and video processing.

Academic Appointments

- **Assistant Professor**, Shiv Nadar University Chennai
  July 2026 – Present
- **Assistant Professor**, Sri Sivasubramaniya Nadar College of Engineering, Chennai
  August 2023 – June 2026
- **Assistant Professor**, SVR Engineering College, JNTUA, Nandyal, Andhra Pradesh
  July 2019 – December 2020
- **Assistant Professor**, BIT Institute of Technology, JNTUA, Hindupur, Andhra Pradesh
  July 2018 – July 2019

Education

Ph.D. in Computer Science and Engineering — January 2021 – June 2024

IIITDM Kancheepuram, Chennai — **CGPA: 9.28**
Thesis: Fault-Tolerant Adder and Multiplier Designs for Mission Critical Systems

Master of Technology in Computer Science and Engineering — October 2018

JNTU College of Engineering, Anantapuram — **CGPA: 7.7**

Bachelor of Technology in Computer Science and Information Technology — June 2013

G. Pullaiah College of Engineering & Technology, Kurnool — **CGPA: 7.1**

Research Areas

- Reconfigurable computing and FPGA-based system design
- Fault-tolerant, and self-healing adder and multiplier architectures
- Approximate computing and significance-driven arithmetic circuits
- Hardware accelerators for artificial intelligence and deep learning
- Hardware architectures for image and video processing

Research Publications

International Journals — **5 publications (SCIE / Scopus indexed)**

[1] Pankaj Kumar Raghuwanshi, Ajay Kumar Reddy Poreddy, Sakali Raghavendra Kumar, and Balasubramanyam Appina, "An Unsupervised Stereoscopic Image Quality Prediction Model Using Perceptual and Statistical Features of Scene Attributes," *IEEE Transactions on Instrumentation and Measurement*, 2025. (Q1, SCIE & Scopus). DOI: 10.1109/TIM.2025.3619656.

[2] Raghavendra Kumar Sakali and Noor Mahammad Sk, "Fault-tolerant multiplier using self-healing technique," *Microelectronics Reliability*, vol. 160, article 115458, 2024. (Q2, SCIE & Scopus). DOI: 10.1016/j.microrel.2024.115458.

[3] Raghavendra Kumar Sakali, Sreehari Veeramachaneni, and Noor Mahammad SK, "Preferential Fault-Tolerance Multiplier Design to Mitigate Soft Errors in FPGAs," *Journal of VLSI Integration*, vol. 93, article 102068, 2023. (Q2, SCIE & Scopus). DOI: 10.1016/j.vlsi.2023.102068.

[4] Sakali Raghavendra Kumar and Noor Mahammad Shak, "Intrinsic Based Self-Healing Adder Design Using Chromosome Reconstruction Algorithm," *Journal of Electronic Testing*, vol. 39, no. 1, pp. 111–122, 2023. (Q3, SCIE & Scopus). DOI: 10.1007/s10836-023-06050-1.

[5] Raghavendra Kumar Sakali, P. Balasubramanian, Ramesh Reddy, Sreehari Veeramachaneni, and Noor Mahammad Sk, "Optimized Fault-Tolerant Adder Design Using Error Analysis," *Journal of Circuits, Systems and Computers*, vol. 32, no. 06, article 2350091, 2022. (Q3, SCIE & Scopus). DOI: 10.1142/S0218126623500913.

International Conferences **4 publications (Scopus indexed)**

[1] Sakali Raghavendra Kumar, Noor Mahammad Sk, P. Balasubramanian, and G. Venkat Reddy, "Novel Embryonic Adder Architecture with Uni-Cellular SelfCheck Unit," in *Proceedings of the 2nd International Conference on Emerging Trends in Engineering*, Atlantis Press, 2023. DOI: 10.2991/978-94-6463-252-1_93.

[2] Sakali Raghavendra Kumar, Sreehari Veeramachaneni, and Sk Noor Mahammad, "Fault-Tolerant Floating-Point Multiplier Design for Mission Critical Systems," in *2024 37th International Conference on VLSI Design and 2024 23rd International Conference on Embedded Systems (VLSID)*, Kolkata, India, January 2024. DOI: 10.1109/VLSID60093.2024.00120.

[3] S. Deepanjali, Noor Mahammad Sk, and Sakali Raghavendra Kumar, "Fault Resilient Micro-Coded Control Unit for Space-Based Digital Systems," in *2024 28th International Symposium on VLSI Design and Test (VDAT)*, Vellore, India, 2024, pp. 1–6. DOI: 10.1109/VDAT63601.2024.10705726.

[4] Sakali Raghavendra Kumar and Sk. Noor Mahammad, "Preferential Fault-Tolerant Based TF32 Floating Point Adder for Mission Critical Systems," in *33rd IEEE Asian Test Symposium (ATS 2024)*, Ahmedabad, India, December 2024. DOI: 10.1109/ATS64447.2024.10915484.

Technical Competencies

Operating Systems	Windows, Linux
Programming Languages	C, Python, JavaScript
Hardware Description Languages	Verilog
FPGA and Design Tools	Xilinx Vivado, Logisim, Dia
Frontend Development	HTML, CSS, React.js, Vue.js
Backend Development	Node.js, Express.js, Spring Boot
Web Services and APIs	RESTful APIs
Database Technologies	MySQL, MongoDB
Development and Productivity Tools	Vite, MS Office, LaTeX, Git

Teaching Portfolio

- Data Structures
- Python Programming
- Computer Organization and Architecture

- Internet Programming
- Digital Principles and System Design
- Object Oriented Programming in Java

Research Supervision and Student Mentoring

- Guided one B.Tech. project team annually from 2023 to 2026.
- Supervised two M.Tech. students in 2024.
- Mentored 25 undergraduate students from the 2024–2028 batch.
- Received a **College Mentor Certificate** in recognition of exceptional mentorship and guidance provided to students participating in the *Bharat AI-SoC Student Challenge (January–March 2026)*, jointly organized by Arm Education and the Chips to Startup Programme, Ministry of Electronics and Information Technology (MeitY), Government of India. The certificate was presented on April 30, 2026.

Academic Service and Professional Membership

- Contributed to consultancy projects, MoUs, departmental research activities, and accreditation processes including NBA, NAAC, IQAC, and ISO.
- Represented the Institution's Innovation Council (IIC) as a member for the CSE Department.
- Served as Class Teacher for two semesters.
- Active member of the Institute of Electrical and Electronics Engineers (IEEE).
- Project Staff in a DRDO-funded project for a two-year tenure; joined the Ph.D. programme under the project category.
- Teaching Assistant for two semesters during the Ph.D. programme (2022–2023).

Academic Events and Programme Organization

- Served as Conference Chair for the IEEE 9th International Conference on Computational Intelligence in Data Science (ICCIDS 2026), January–March 2026, coordinating with global and local committees to secure approvals and support event execution. Website: ICCIDS 2026 website. Conference details: IEEE conference record.
- Received a USD 3,250 sponsorship grant from the IEEE Signal Processing Society to organize the 2025 IEEE SPS Summer School, "Multimedia Signal Processing and Analysis: Bridging Theory and Practice Using Computational Intelligence." Co-organized the international summer school, including expert lectures, hands-on sessions, and interactive discussions.
- Organized an Industry-Oriented Course on "Dockerization and Deployment of Web Applications," February 27–28, 2026.
- Organized a Value-Added Course on "User Interface & User Experience Design," February 16–March 2, 2026.
- Organized a workshop on "Parallel Processors and Its Challenges" (partial content delivery), September 29, 2025.
- Organized a Value-Added Course on "Full-Stack Web Application Development using Spring Boot & Vue.js," September 22–October 15, 2025.
- Organized a workshop on "Parallel Processors and Its Challenges," March 20, 2025.
- Organized a workshop on "Open-Source Resourcing Tools," March 21, 2025.

Certifications

- NPTEL Certification in Cloud Computing with a score of 76% (Elite + Silver).
- HackerRank Certification in C (Basic) Skill Test.

Research Vision

My future research aims to advance approximate datapath units for machine-learning and deep-learning accelerators, building on my doctoral work on reliable adder and multiplier design. The focus is on energy-efficient, high-performance hardware architectures for image and video processing. I plan to explore FPGA-based accelerators that exploit reconfigurability, parallelism, and application-specific customization to balance accuracy, speed, and power, particularly for resource-constrained edge-computing environments. The broader goal is to develop scalable, low-power accelerators for real-time AI-driven vision systems.

References

Dr. Noor Mahammad Sk
Associate Professor
HPRCSE Lab, PEMS Block
IIITDM Kancheepuram, Chennai – 600127
Email: noor@iiitdm.ac.in
Contact: 9176010587

Prof. Masilamani V
Professor
Room: 108-B, Annexure Building
IIITDM Kancheepuram, Chennai – 600127
Email: masila@iiitdm.ac.in
Contact: 044-27476346

Declaration

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Place: Chennai

Date: November 9, 2026 **(Dr. Sakali Raghavendra Kumar)**